

SYSTEM ANALYSIS and DESIGN: THE REVOLUTIONS and THE TRANSITION

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Abstract- This academic paper aims to reveal the history of system analysis and design as a story of inventions and innovations. We seem to go back to our past, the middle of the 20's century, this was the time when the whole world was in the transition period to a modern computer time. Today is a day of rapid progress in the development of technology, thus, the primary role of System Analysis and Design (SAD) is one of the significant matters, especially in the industry of Information Systems (IS). SAD which is also known as Systems Development Life Cycle (SDLC) builds both simple and complex systems which are the foundation of the development process. This paper contains all the components necessary for a highly motivational description of our times where SAD emerges as a key factor in the development of technological progress.

Keywords– System Analysis, Information System, System Development Life Cycle, System Design, Technology, Revolution of Technology

I. INTRODUCTION

SAD, which stands for system analysis and design, is a methodology that is applied in information systems for business. It is a combination of IT Infrastructure, people, and data that formulate top-notch systems designed to specifically satisfy the needs of each business. System analysis and design are familiar with the System Development Life Cycle

(SDLC). The purpose of SDLC acts as a guideline for the developer to design and develop the information systems. According to Shirley Radack [1], several SDLC models and methodologies can be used, for each of them includes several steps and phases. The basic steps that are always used in system development as in Figure 1 [2] consist of planning, design, analysis, implementation, and maintenance.

For each project, all the phases are essential to achieving the right result. In this preparation stage, the team identifies the project objectives, targets, and the project scope. The phase of analysis will include the study of the present system, the identification of issues, and the enumeration of parameters of the new system. The design phase requires the team to draft a plan for the new system and build a development blueprint that describes each step. During the implementation stage, the system is subject to building and testing begins and proceeds until it satisfies the requirements. But the last role is the maintenance one: it refers to verifying that the system keeps functioning properly and tackles any problems that appear.

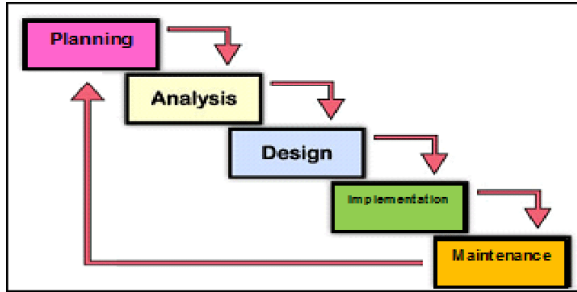


Fig. 1 Traditional waterfall of SDLC

II. IMPORTANCE OF SYSTEM ANALYSIS DESIGN

The importance of system analysis design was taken from “What is system analysis and design? (Plus benefits)” [3]:

1) Cost Reduction and Resource Optimization

The more efficient the systems are, the more it will lead to cost savings by eliminating some of the inefficiencies. Resources that are saved can be allocated to other critical areas within the organization.

2) Improved Operational Efficiency

One of the main purposes of SAD is to help organizations enhance their systems or create a new system to overcome the older one. By optimizing processes and reducing redundancy, the business will achieve higher operational efficiency.

3) Boosting user efficiency and contentment

Well-designed systems and minimized manual effort can lead to increased productivity and cost savings for users to use the system.

III. HISTORY OF SYSTEM ANALYSIS AND DESIGN

A. Early Years (1940s-1950s)

The evolution of system analysis and design began with the development of the first electronic computer, ENIAC. As said in the ENIAC journal article [4], ENIAC was created due to the huge

demand of war to complete tasks beyond human capabilities, such as computing the ballistic firing tables in World War II. In this early era of technology, everything was being done manually by the work of humans. Thus, with the existence of the first-world computer, most of the manual business processes started to transition into an automated environment as it was difficult for a human to monitor all the processes and manage them efficiently. The role of system analysis and design during this time was to improve the efficiency and accuracy of data processing tasks.

B. Structured System Analysis and Design (1960s-1970s)

As technology developed, many companies found out that designing diverse data infrastructures for diverse applications was very expensive. During the early 1960s [5], a Database Management System (DBMS) was developed to manage bill of materials transactions processing (BOMP). When DBMS is used with the SQL database language, the data is organized into tables made up of columns and rows, and then rules (constraints) are set to keep everything organized [6]. Most of the early DBMS models were based on a network structure except for IBM’s IMS which uses a hierarchical structure. As computer technology advances, the idea of applying it across the enterprise level to automate business processes results in the Management Information Systems (MIS) era. Data Management (DM) organizations emerged to eliminate redundancies and improve system integration to achieve consistent data results. However, many of these organizations are too focused on the DBMS and begin to lose sight of their original goals, resulting in data redundancy and missed opportunities to segment and utilize data. In

the 1970s, many new treatments were developed, including those by MBA, John Toellner, and Atlantic Software.

C. Object-Oriented Analysis and Design (1980s-1990s)

In the late 1980s and 1990s, Object-Oriented Analysis and Design (OOAD) was widely accepted as a technical procedure to manipulate business, application, or system models [7]. The analysis methodology for this process requires a collection of works of the current system model, to determine the new requirements of the system. Object-oriented programming and different types of data modeling are mainly used for this process depending on the cases. The basic idea of this process is to make the software design and development simpler by classifying all of the components into objects, classes, and methods. By doing this, it can help the developer to manage the complexity of the project by breaking the system into smaller and simpler parts.

There are four fundamental steps in OOAD which are Function Modelling Techniques, Finding and identifying the business objects, organizing the objects and identifying their relationships, and modeling the behavior of the objects. The advantages of using this procedure can make the system more reliable and flexible and also may reduce the maintenance cost as all of the processes are being encapsulated, making the existing model reusable for new improvements and developments.

D. Agile and Iterative Approaches (2000s-Present)

We have passed through almost 60 years since the first revolution or transition of technology which started in the early 1960s. The biggest change

that we can notice is our “gadgets” have become smaller, faster in processing information and data, and even larger in capacity. The least capacity for phones nowadays I believe is 64 GB which is humongous compared to the older ones. Most of the devices that have been created nowadays such as all devices that interface with the PC, smartphones, PlayStation, and digital cameras are aimed at either communications or entertainment, giving us greater mobility, and even maybe causing us a little bit of a lack of social interaction which is one of the main downsides of gadgets.

As we already know, the rapid changes in technology have created an urge and demand to build software faster than the ones that already exist in the technology industry. This also could be due to many factors such as competitive markets, customer expectations, and the need for businesses to improve their services quickly. However, while speed is important, it's also essential to maintain a high standard of quality, security, and user experience in software development. So, to achieve these expectations, we can't use the traditional “full-bodied” methodologies such as the Waterfall model due to being rigid and slow, making them less effective for today's fast-paced revolution. Instead of using this traditional approach, “Agile” and “Extreme Programming” are the best choices because they're flexible, iterative approaches that emphasize customer collaboration and feedback from customers and small, rapid releases. This methodology aims to reduce inefficiencies and improve software quality by quickly adapting to changes.

IV. CONCLUSIONS

System analysis and design are very crucial to ensure the software is of good quality, user-friendly, and safe. As technology rapidly expands, challenges also come with it. This requires a combination of good skills and expertise to keep up with the new trends and to maintain the software always in good shape of quality.

In conclusion, system analysis and design is like a journey from doing things by hand to using computers. It's mainly about improving data management and keeping up with people's demands. It is really important to quickly adapt and respond in today's fast-paced digital world.

ACKNOWLEDGEMENT

A special thanks to our lecturer, Puan Aryati Binti Bakri, who acted as our guide and supervisor in developing this academic paper. She is the one to be thanked and appreciated for making this academic work possible.

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